

- 2 (a) Sucrose is a disaccharide.

Fig. 2.1 shows how sucrose is broken down in a reaction with hydrochloric acid.

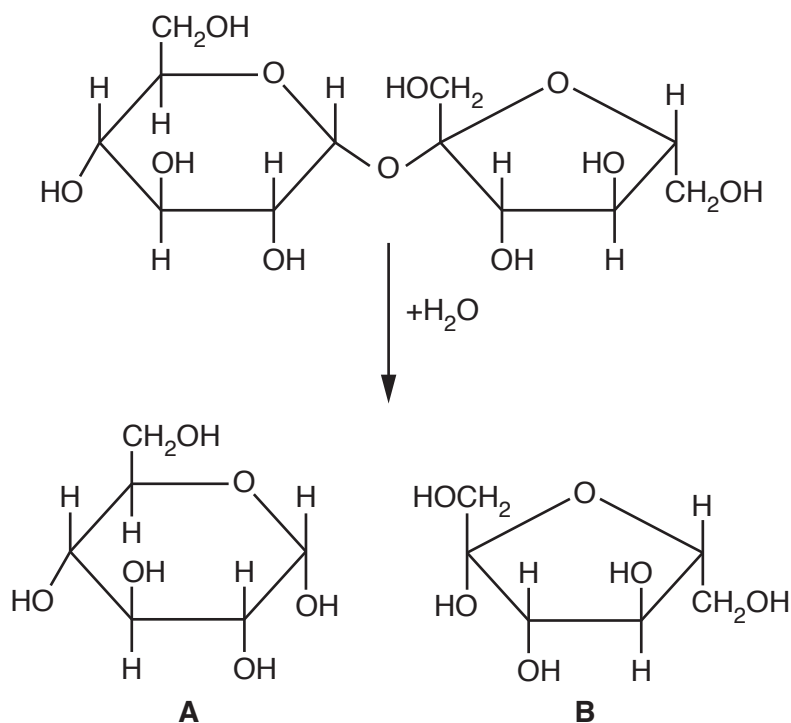


Fig. 2.1

- (i) Name the products, **A** and **B**, of the reaction shown in Fig. 2.1.

A

B

[2]

- (ii) Name the type of bond that is broken in the reaction shown in Fig. 2.1.

.....[1]

- (iii) State the type of reaction shown in Fig. 2.1.

.....[1]

- (b) When Benedict's solution is added to a sucrose solution and put into a boiling water-bath, no change in colour is observed.

State why no colour change is observed.

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- (c) Sugar beet plants, *Beta vulgaris*, store sucrose in their roots.

Describe **and** explain how sucrose is transported from the phloem in the leaves of sugar beet to storage tissues in the root.

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[Total: 10]